BUILD AI-AMD

AI (Artificial Intelligence)

Making machine smart enough to perform task that require human intelligence.

Example: Self-driving car, Chat-GPT, Smart washing machine.

ML (Machine Learning)

A part of AI, where machine learns form data and make predictions.

Example: An AI model that predict whether the review is fake or genuine on the e-commerce website.

DL (Deep Learning)

A part of ML, that uses Neural Network to learn complex pattern from data.

Example: CNN (Convolutional Neural Network) used for image classification.

Prompt Engineering

Prompt → Input

Engineering → The work done by engineer

Prompt engineering means giving clear instructions to an AI so that it gives you the answer you want.

Example: Suppose you ask an AI:

❌ Bad prompt:
'Tell me about AI.'

The AI might give you a very long and complicated answer.

✅ Better prompt:
'Explain Artificial Intelligence in simple English with two real-life examples.'

Now the AI knows exactly what you want and can give you a more useful answer.

Real-life example: Imagine you are talking to a chef.

- You say, 'Make something.' The chef doesn't know what you want.
- You say, 'Make spicy egg fried rice for one person.' Now the chef knows what to prepare.

Prompt engineering works in the same way. You give the AI clear instructions about what you need.

LLM (Large Language Model) As Predictive Engine

LLM: Large Language Model, an AI model trained on a large amount of **Text/Data**.

Predictive: Guessing what is likely to come next based on available information.

Engine: A system that performs a task.

How does an LLM work?

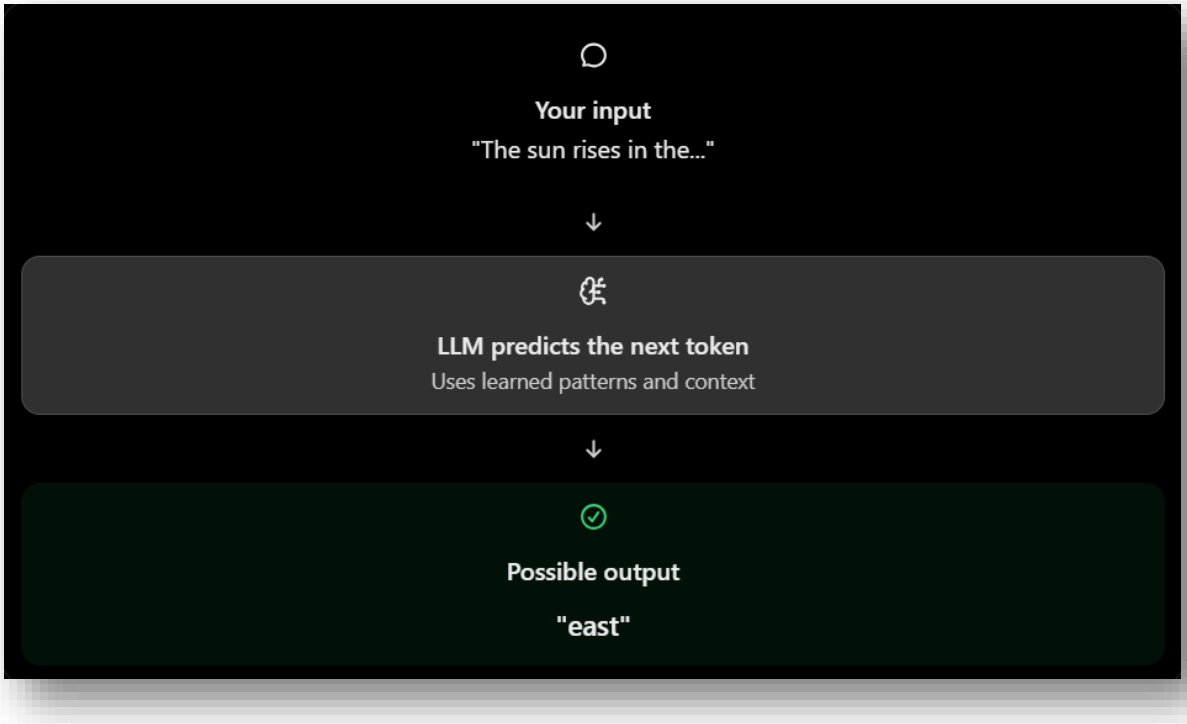
Imagine you write this sentence:

'The sun rises in the ___.'

What word do you expect next?

Probably 'east.'

An LLM works in a similar way. It learns patterns from a huge amount of text during training and uses those patterns to predict likely next tokens.



The model then predicts another token, and another, until it produces a complete response.

How does it generate a full answer?

Suppose you ask:

‘What is artificial intelligence?’

The LLM generates a response piece by piece:

1. Artificial
2. Intelligence
3. is
4. a
5. technology
6. that ...

The model repeatedly predicts the next token using the previous context. This is how it creates sentences and paragraphs.

Tokenization

Tokenization means, the input/prompt you give to the AI is broken down into small words, so that a computer can process and analyze language.

Example: If you tell Chat-GPT that ‘What is AI’

1. It will Broke it into tokens (Words).
2. Convert that token into numbers, because computer understand Binary not English.
3. Then it will see the meaning of each word.
4. Then it will see the relationship between the words.
5. Then based on the pattern it has learned it will give the output or it will predict the next token, until the whole response is generated.

Controlling Token Length

Token length controls how long the AI’s response can be.

- Low token limit: Short answer.
- High token limit: Longer, more detailed answer.

Example:

- Limit = 20 tokens → A brief explanation.
- Limit = 200 tokens → A more detailed explanation.

Controlling Temperature (Randomness)

Temperature controls how predictable the AI's responses are.

- Low temperature (e.g., 0.2): More predictable and consistent answers.
- High temperature (e.g., 1.0): More varied and creative answers.

Example: If you ask an AI to name a business

- Low temperature → It may suggest a straightforward name like 'Tech Solutions.'
- High temperature → It may suggest a creative name like 'Pixel Galaxy.'

Remember:

Token length = How much the AI generates.

Temperature = How varied the AI's choices can be.

Top-K Sampling

Top-K means the AI considers only the K most likely next tokens and chooses from them.

Example: Suppose the AI is completing

'I like to eat...'

Possible next tokens:

Token	**Probability**
Pizza	40%
Rice	30%
Burger	20%
Stones	10%

If Top-K = 2, the AI considers only Pizza and Rice.

It ignores the other options during selection.

Top-P Sampling

Top-P means the AI considers the smallest group of likely tokens whose combined probability reaches a chosen threshold.

Suppose Top-P = 0.7 (70%).

- Pizza = 40%
- Rice = 30%

Remember: Top-K and Top-P are sampling parameters used during LLM text generation to control the selection of the next token. Not every AI platform or model supports both settings.

Types Of Prompting

1. **Zero-Shot Prompting**: Zero-shot prompting means asking an AI to perform a task without giving it any examples.
   Example: Prompt:
   'Classify this review as Positive or Negative: This movie was amazing!'
   AI Output: Positive
   Here, you didn't give the AI any examples of how to classify reviews. You simply gave it the task.

2. **One-shot Prompting**: One-shot prompting means providing one example before asking the AI to perform ɑ similar task.
   Example: Review: 'The movie was fantastic.'
   Label: Positive
   Now classify this review: 'The movie was boring.'
   Label: The first example demonstrates the expected pattern. The model uses that pattern to classify the new review.

3. **Few-shot Prompting**: Few-shot prompting means providing multiple examples in ɑ prompt so that the model can follow the demonstrated pattern.
   Example: Pizza Order Conversion to JSON
   Example 1: Order: One small cheese pizza.
   Output: {'size': 'small', 'topping': 'cheese'}
   Example 2: Order: One large mushroom pizza.
   Output: {'size': 'large', 'topping': 'mushroom'}
   Now convert this order: Order: One medium paneer pizza.
   Expected output: {'size': 'medium', 'topping': 'paneer'}

4. **System Prompting**: System prompting means setting the overall rules and behaviour of an AI.
   Example:
   'You must always explain topics in simple English. Keep your answers short.'
   The AI follows these instructions while responding.
   Remember: System prompt = Rules for the AI.

5. **Contextual Prompting:** Contextual prompting means giving the AI background information so it can answer appropriately.
   Example: 'I am a BSc second-year student learning AI. Explain machine learning according to my level.'
   The AI uses this background to make the explanation suitable for you.

Remember: Contextual prompt = Background information.

6. **Role Prompting:** Role prompting means asking the AI to act as a particular person or expert.
   Example: 'Act as an experienced AI teacher and explain neural networks with real-life examples.'
   The AI responds in a teaching style based on the role you requested.
   Remember: Role prompt = Who the AI should act as.

Tone and Style in Prompt Engineering

Tone and style control how the AI communicates its answer.

1. Formal vs Informal

Formal: Professional and polite language.

- Prompt: 'Explain artificial intelligence in a formal tone."'
- Output: 'Artificial intelligence is a technology that enables machines to perform tasks that typically require human intelligence.'

Informal: Casual and friendly language.

- Prompt: 'Explain artificial intelligence like you're talking to a friend.'
- Output: 'AI is basically technology that helps computers do smart things, like answering questions!'

2. Direct vs Descriptive

Direct: Gives the answer clearly and briefly, without extra details.

- Prompt: 'What is AI? Answer directly in one sentence.'
- Output: 'AI enables machines to perform tasks that normally require human intelligence.'

Descriptive: Gives more details, explanations, and examples.

- Prompt: 'Explain AI descriptively with examples.'
- Output: 'Artificial intelligence (AI) enables machines to perform tasks that normally require human intelligence. For example, AI can recognize faces, understand speech, recommend movies, and answer questions.'

Step-Back Prompting

Step-back prompting means asking a general question first before asking the specific question.

It helps the AI understand the bigger picture before solving a particular problem.

Example: Suppose you ask an AI

Direct question:
'Why do I fail in exam?'

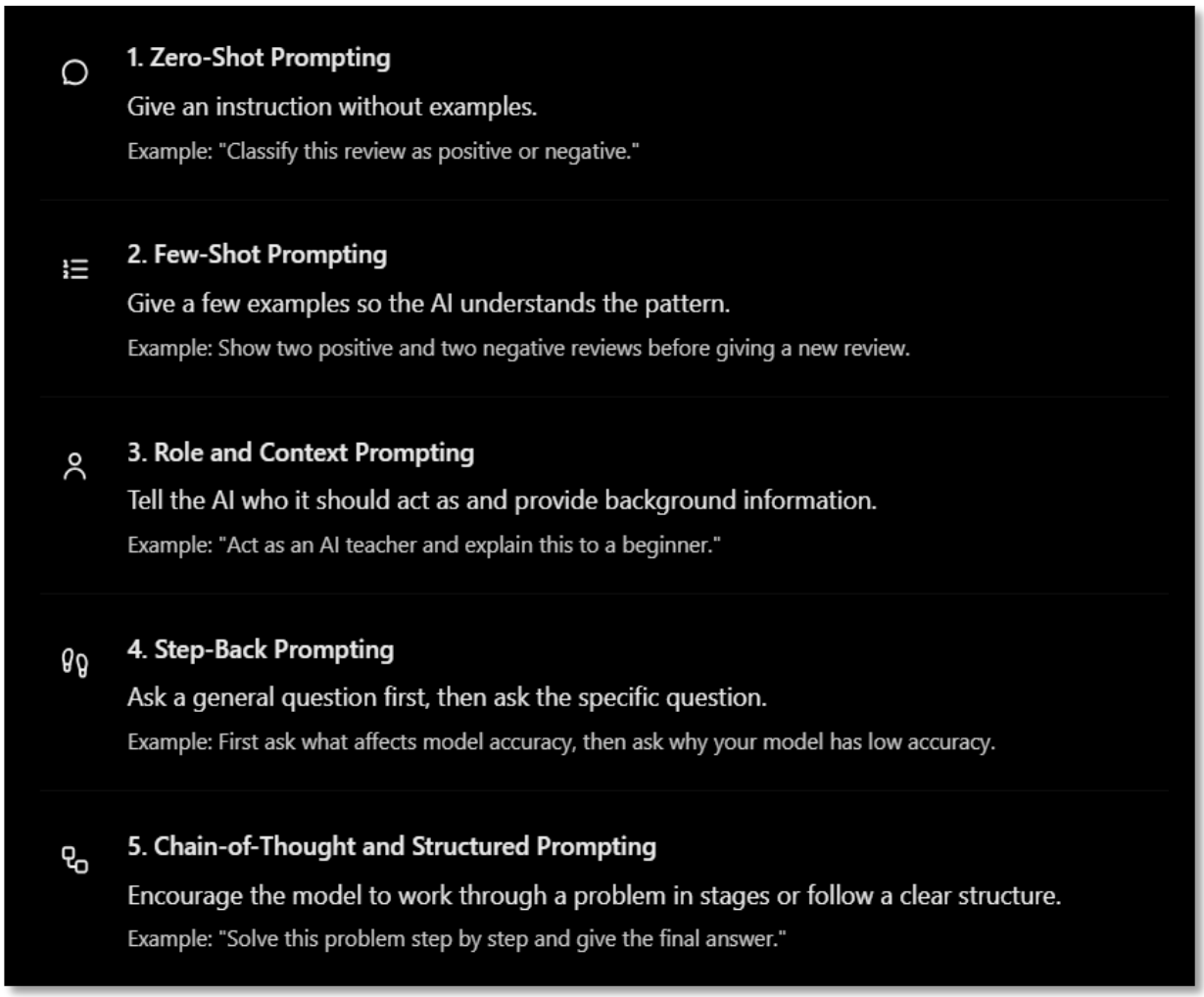
With step-back prompting, you first ask:

- Step 1 -- General question: 'What are the common reasons students fail in exam?'
- Step 2 -- Specific question: 'Based on these reasons tell me why do I fail in exam.'

The AI first considers the general concepts, then applies them to your specific problem.

Evolution of Prompting Techniques

The evolution of prompting techniques means how the way we give instructions to AI has improved over time to get better answers.



Chain-of-Thought (CoT)

You ask the AI to solve a problem step by step.

Example prompt:

‘What is 25 × 12? Think through the problem step by step.’

The AI works through the calculation:

- 25 × 10 = 250
- 25 × 2 = 50
- 250 + 50 = 300

Here, you tell the model to approach the problem step by step.

Automatic Chain of Thought (Auto-CoT)

Auto-CoT is a prompting technique in which an AI automatically generates reasoning examples to help solve similar problems.

Let’s understand it simply.

Example: Suppose you want an AI to solve 100 mathematics questions.

- Normal prompting: Ask each question and get an answer.
- Manual Chain of Thought: You provide examples showing how to solve questions step by step.
- Automatic Chain of Thought (Auto-CoT): The system automatically creates reasoning examples for selected questions and uses them to help solve other questions.

Suppose you have 1,000 maths questions.

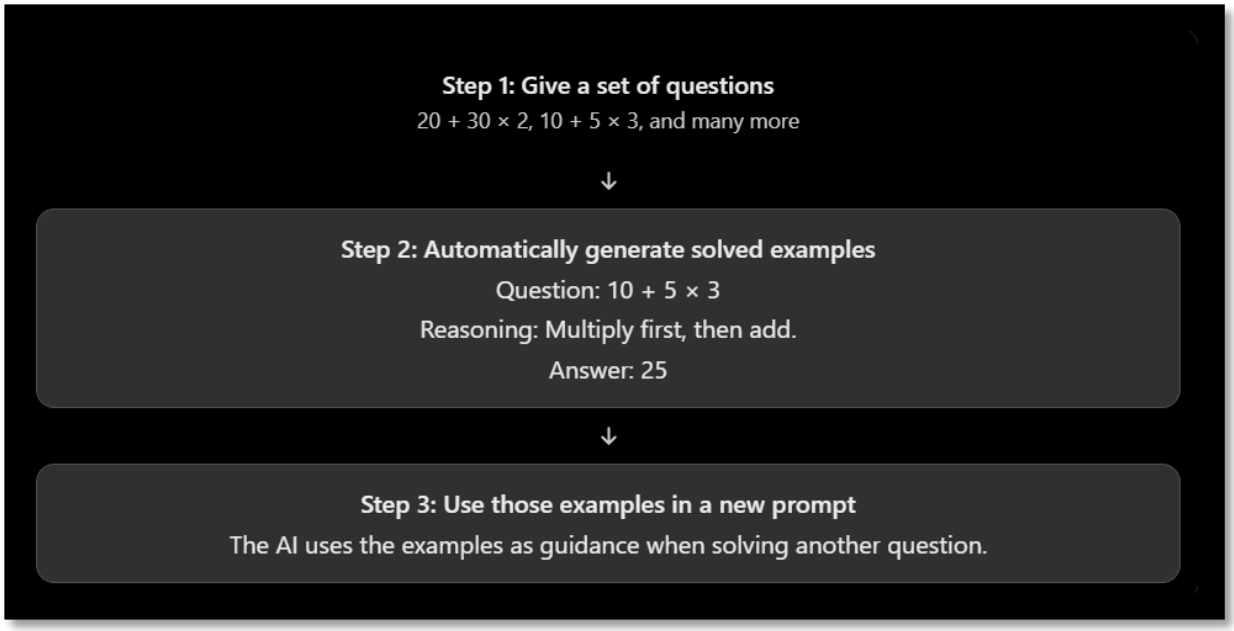
Option 1: Use CoT

You ask the AI to solve each question step by step.

- Question 1 → Solve step by step.
- Question 2 → Solve step by step.
- Question 3 → Solve step by step.
- And so on.

Option 2: Use Auto-CoT

- The system selects representative questions.
- The AI automatically generates step-by-step examples for them.
- These examples are reused in prompts to guide the answers to other questions.



Tree of Thoughts (ToT)

Tree of Thoughts is a prompting technique in which AI explores multiple possible ways to solve a problem instead of following only one path.

Let's understand with an example.

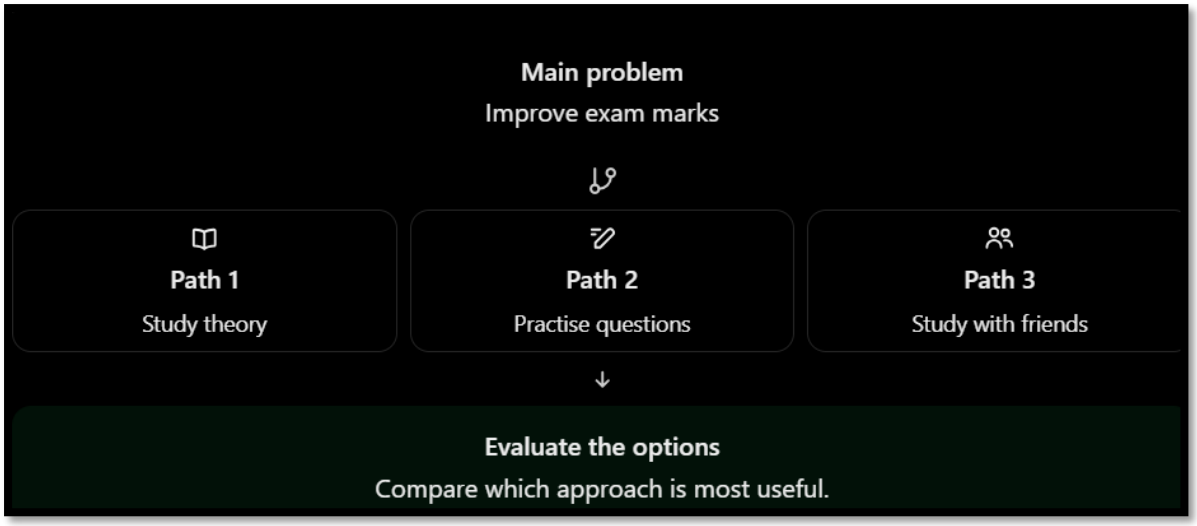
Example: You want to solve a difficult problem.

Suppose you ask AI: 'How can I improve my exam marks?'

With Chain of Thought (CoT), AI follows one reasoning path:

Study daily → Practise questions → Revise → Improve marks.

With Tree of Thoughts (ToT), AI explores different paths:



How to achieve Tree of Thoughts (ToT) in ChatGPT

I need to prepare for an AI exam in 5 days.

- Use the Tree of Thoughts approach:
- Generate three different study plans.
- Evaluate each plan based on time, difficulty, and expected usefulness.
- Identify the strongest plan.
- If a plan has weaknesses, explore alternatives.
- Recommend the best final plan and explain why.

What happens?

- ChatGPT explores three possible plans.
- It compares their strengths and weaknesses.
- It selects the most promising plan and explains its recommendation.

Self-Consistency

The AI generates multiple independent solutions to the same problem and compares their final answers.

- Path 1: 15 x 12 = 15 x (10 + 2) = 180
- Path 2: 15 x 6 x 2 = 180
- Path 3: 10 x 12 + 5 x 12 = 180

All three give 180, so the answer with the strongest agreement is 180.

Main idea: Generate multiple answers and choose the answer that appears most consistently.

Graph of Thoughts (GoT)

Graph of Thoughts allows an AI to explore ideas, connect them, and combine information from different reasoning paths.

Imagine you ask: ‘Where should I go to picnic with class.’

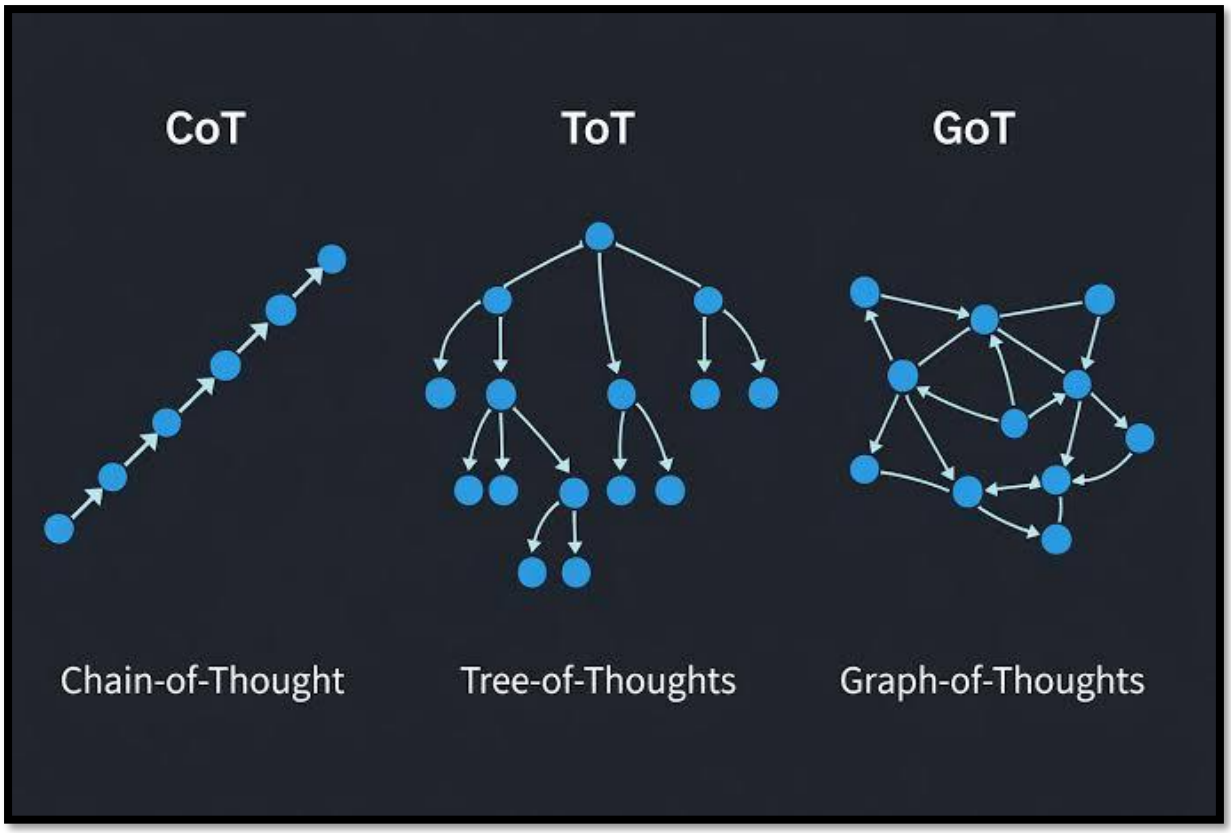
It will explore Budget, Location, Fun.

Then it will connect the all ideas or paths and gives the output.

Example: ‘If you ask AI I am starting a Business, tell me the idea to develop a business.’

AI will explore Product, Customer, Budget, Marketing.

Then AI will combine all these and develop a Business Plan.



Skeleton of Thought (SoT)

Skeleton of Thought is a technique designed to generate answers faster by creating the main outline first and then expanding its points.

Imagine you ask: ‘How can I organize a birthday party?’

Step 1: AI creates a skeleton (outline)

1. Choose a budget.
2. Select a venue.
3. Invite guests.
4. Arrange food and decorations.
5. Plan activities.

Step 2: AI expands the outline

1. Budget: Decide how much money you can spend.
2. Venue: Choose your home, a restaurant, or a party hall.
3. Guests: Prepare a guest list and send invitations.
4. Food and decorations: Arrange a cake, snacks, balloons, and lights.
5. Activities: Plan games, music, and other entertainment.

ReAct Prompting (Reasoning + Acting)

A technique in which AI thinks about what to do, then takes an action, then observe the result and then decide what to do next.

Example 1: You ask Chat-GPT, what is the weather in Chennai Today?

1. Reasoning – Think → I need current weather info.
2. Acting – Take Action → Use weather tool to check condition.
3. Observation – See the result → Tool returns weather condition.
4. Final Answer.

Example 2: Find a laptop under the budget 50k.

1. Find a laptop within Budget.
2. Search the shopping website.
3. Check the product and price.
4. Recommend the laptop.

Automatic Prompt Engineering

Using AI to automatically create or generate prompts instead of writing all the prompts yourself.

Example: Imagine you want AI to classify whether the movie review is positive or negative?

1. You give task.
2. AI generate prompts like
    a. Prompt A: Is this review positive or negative?
    b. Prompt B: Read the review and classify its sentiment as Positive or Negative.
3. Test the prompt and compare them.
4. Select the one which gives the best result.

Emotion Prompting

Adding emotional or motivational words to a prompt to encourage AI to give more careful example.

Example: Normally → Explain AI.

With Emotion → Explain AI carefully. This is very Important for my exam I really need to understand.

Code Prompting

Giving AI instructions to write, explain, or improve the code.

Example: You tell Chat-GPT to write a Python Program that add 2 Numbers.

OR, Find the error in the code and provide correct version.

print(Hello World) → ✖

print("Hello World") → Correct